2026

Bases de dades no relacionals

TASK FOR BD07

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ENLACE

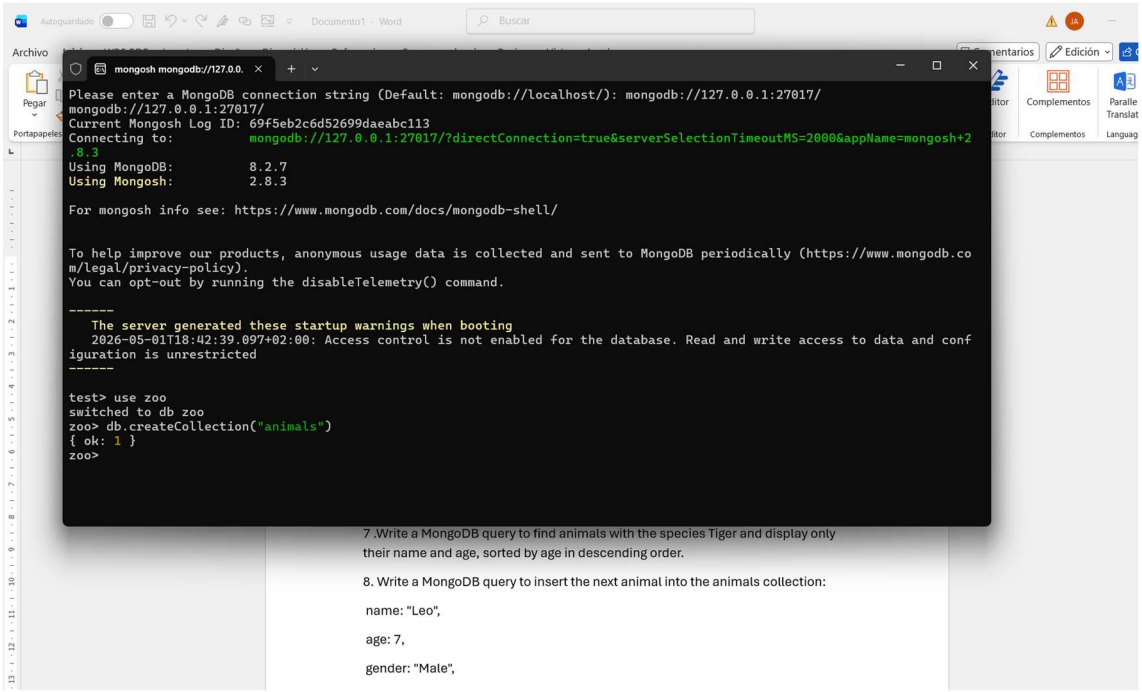
https://github.com/Deya2609/Base-de-datos/blob/main/Task7.%20Mongo%20db

VIDEO LOOM

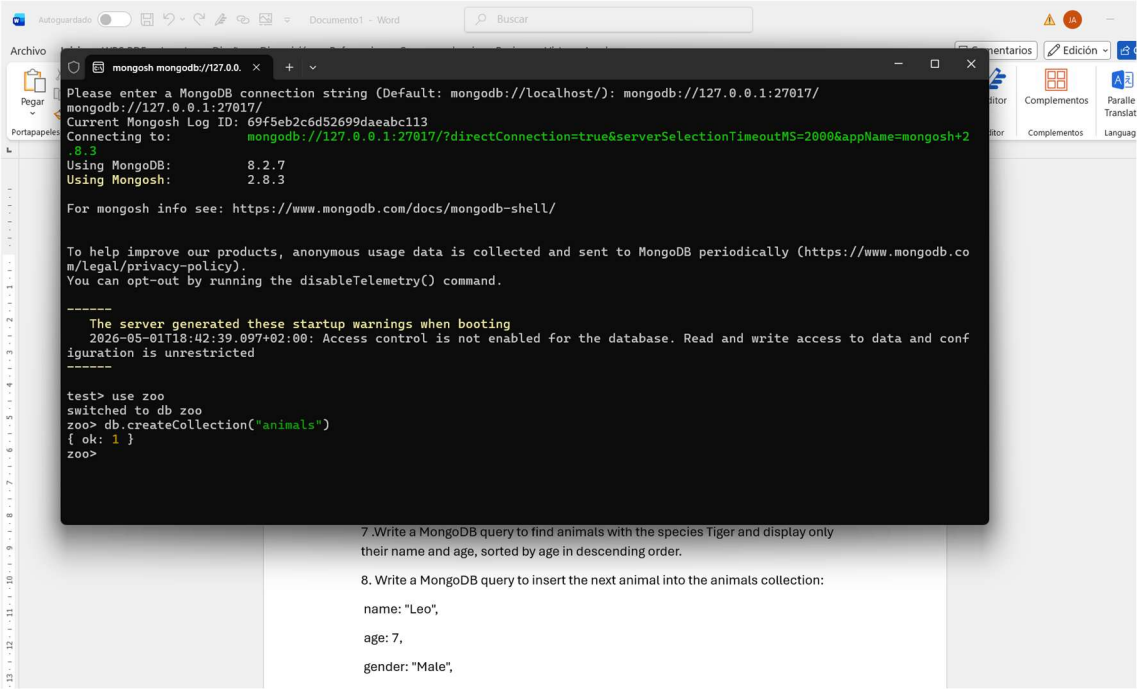
https://www.loom.com/share/f49d17b67dfa4ab7bdcb7a28c8f0e325

Write the code used to perform the following tasks from the MongoDB shell:

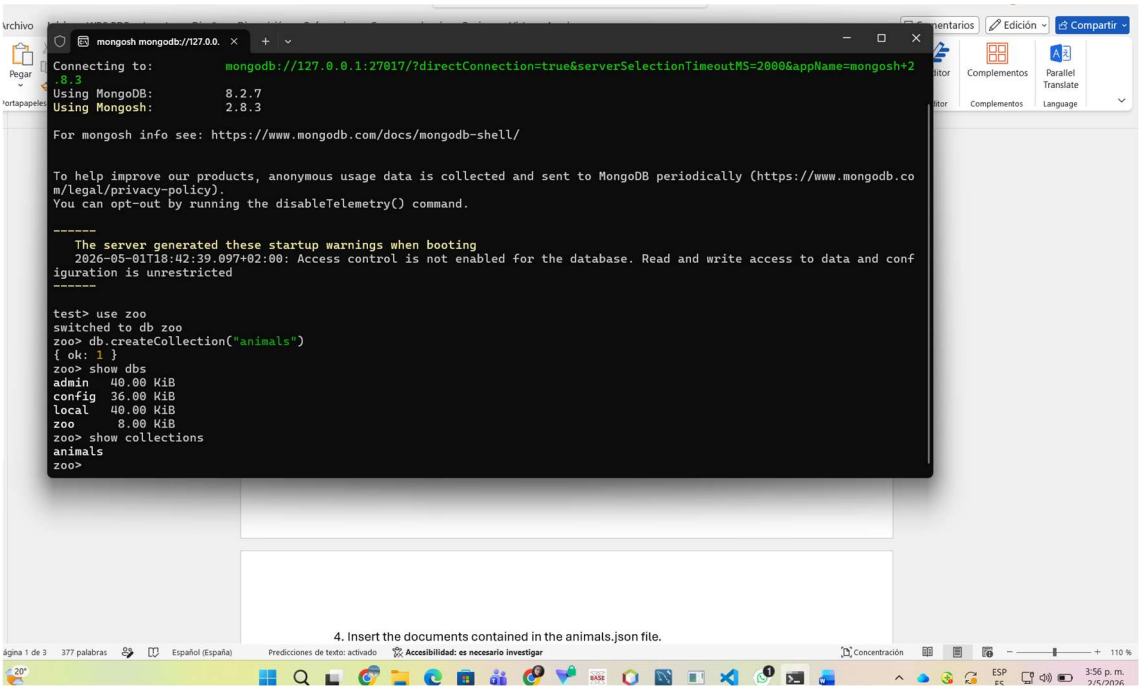
1. Create a database called zoo.



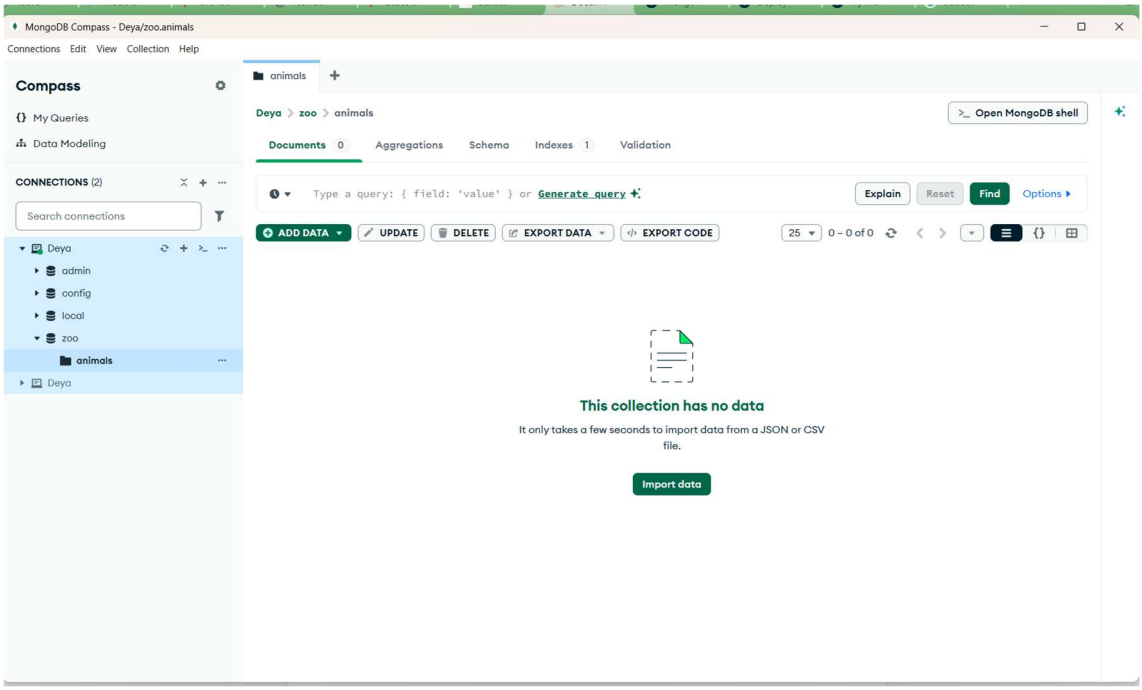
2. Insert a collection called animals.

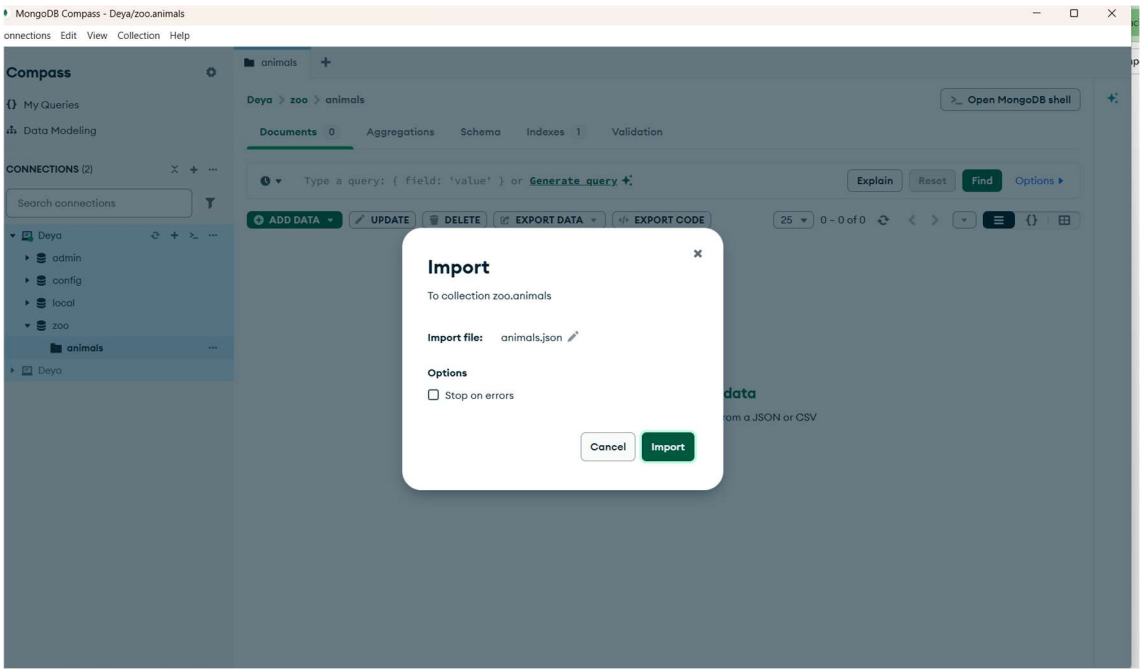


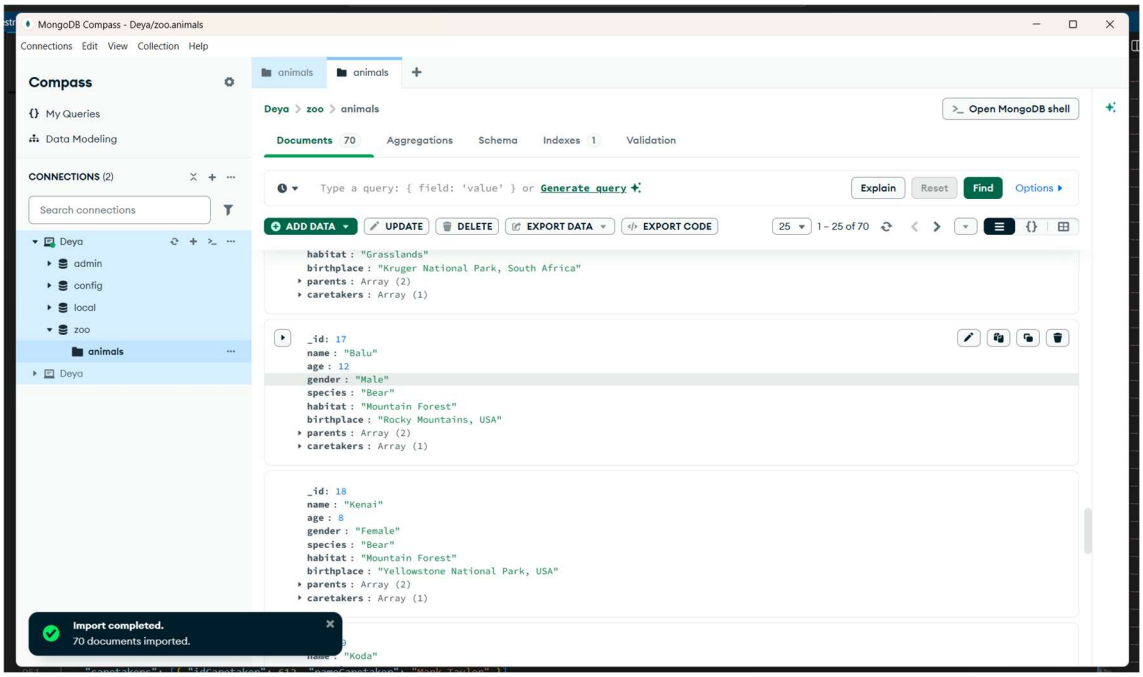
3.Confirm that the database and the collection have been successfully created.



4. Insert the documents contained in the animals.json file.

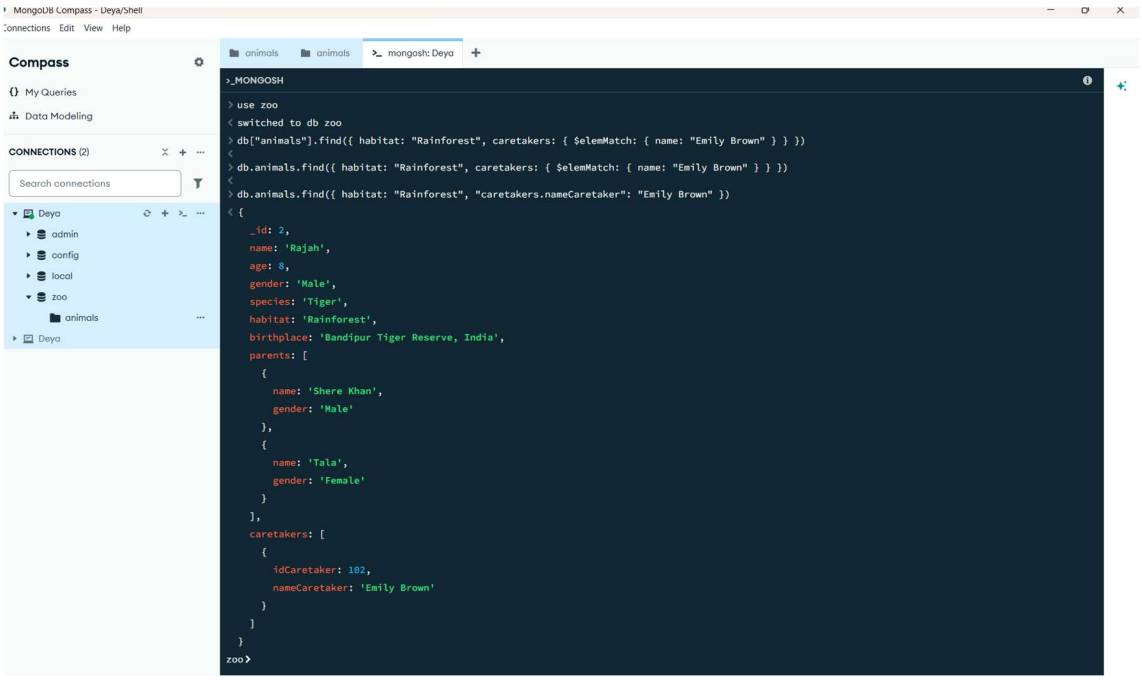

MongoDB Compass - Deya/zoo.animals
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (2)
Search connections
Deya
admin
config
local
zoo
animals
Deya
animals
Deya > zoo > animals
Open MongoDB shell
Documents 0
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
0 – 0 of 0
This collection has no data
It only takes a few seconds to import data from a JSON or CSV file.
Import data


MongoDB Compass - Deya/zoo.animals
Import
To collection zoo.animals
Import file: animals.json
Options
Stop on errors
Cancel
Import



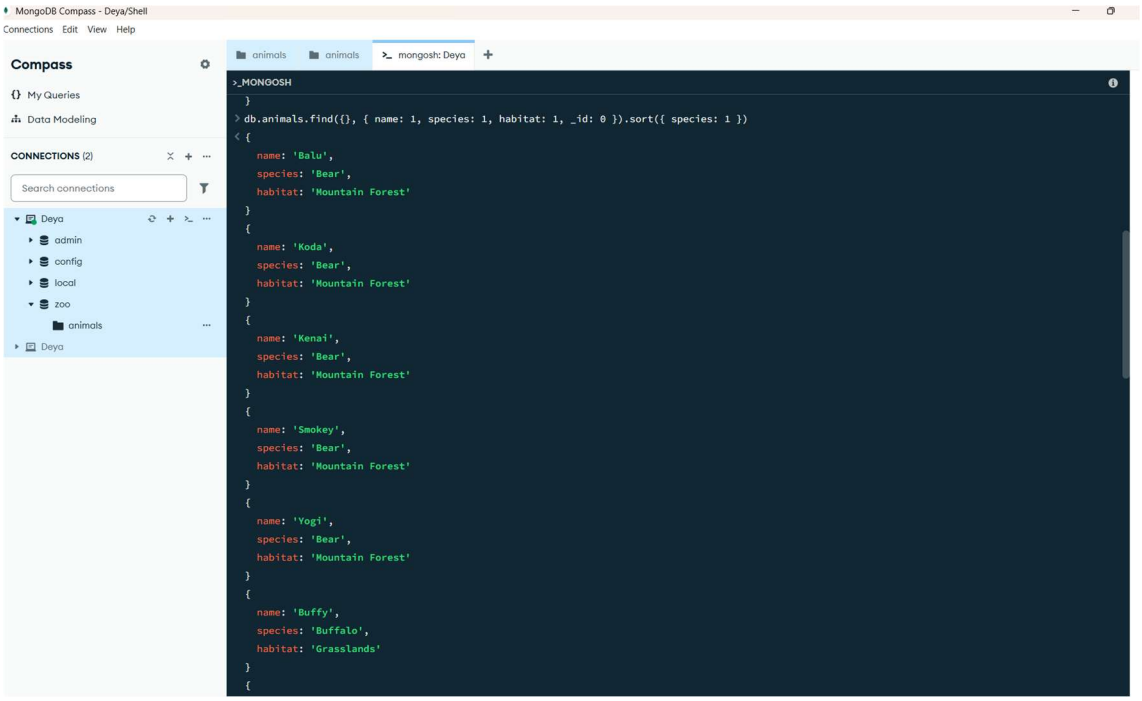
5- Write a MongoDB query to find animals that have Rainforest as their habitat and are attended by a caretaker named Emily Brown.

db.animals.find({ habitat: "Rainforest", "caretakers.nameCaretaker": "Emily Brown" })



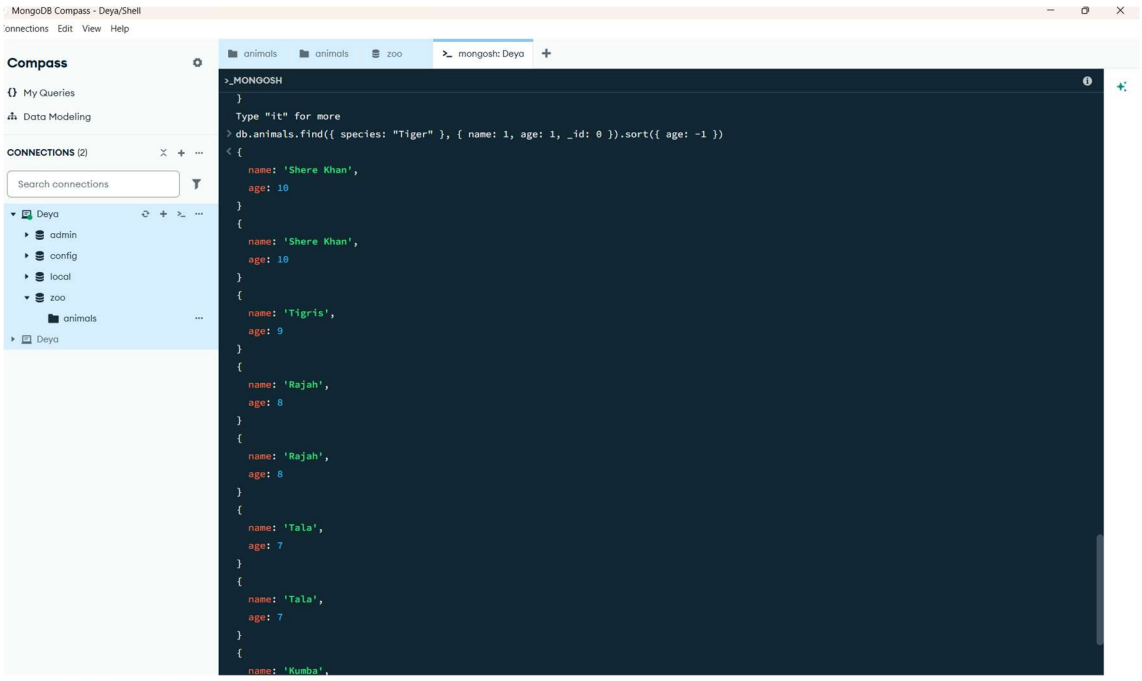
6- Write a MongoDB query to display only the name, species, and habitat of all animals sorted by their species in ascending order.

db.animals.find({}, { name: 1, species: 1, habitat: 1, _id: 0 }).sort({ species: 1 })



7 .Write a MongoDB query to find animals with the species Tiger and display only their name and age, sorted by age in descending order.

db.animals.find({ species: "Tiger" }, { name: 1, age: 1, _id: 0 }).sort({ age: -1 })



8. Write a MongoDB query to insert the next animal into the animals collection:

name: "Leo",

age: 7,

gender: "Male",

species: "Lion",

habitat: "Savanna",

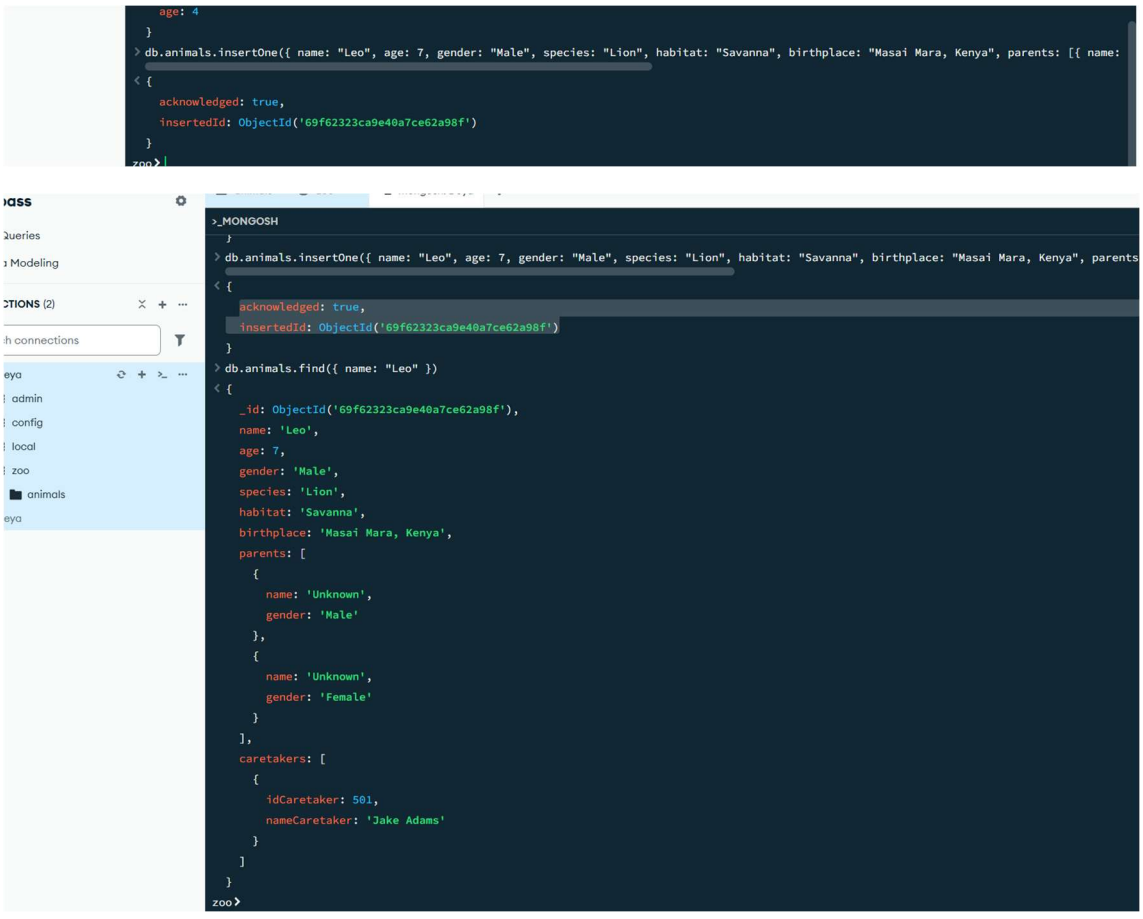
birthplace: "Masai Mara, Kenya",

parents: "Unknown", gender: "Male", Unknown", gender: "Female"

caretakers: id: 501, name: "Jake Adams" }]

db.animals.insertOne({ name: "Leo", age: 7, gender: "Male", species: "Lion", habitat: "Savanna", birthplace: "Masai Mara, Kenya", parents: [{ name: "Unknown", gender: "Male" }, { name: "Unknown", gender: "Female" }], caretakers: [{ idCaretaker: 501, nameCaretaker: "Jake Adams" }] })

db.animals.find({ name: "Leo" })

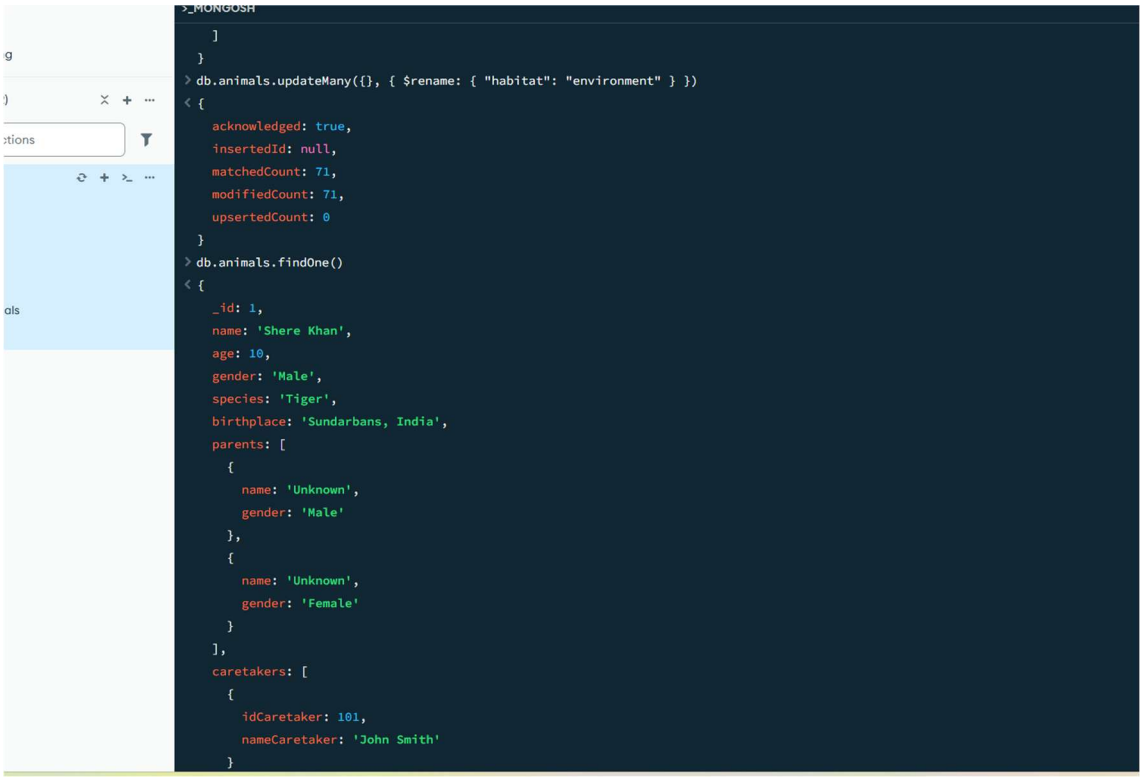


9. Write a MongoDB query to update the age (to 8) of the animal you just inserted using the $set operator

db.animals.updateone({name: “Leo”}, {$set: {age: 8}}

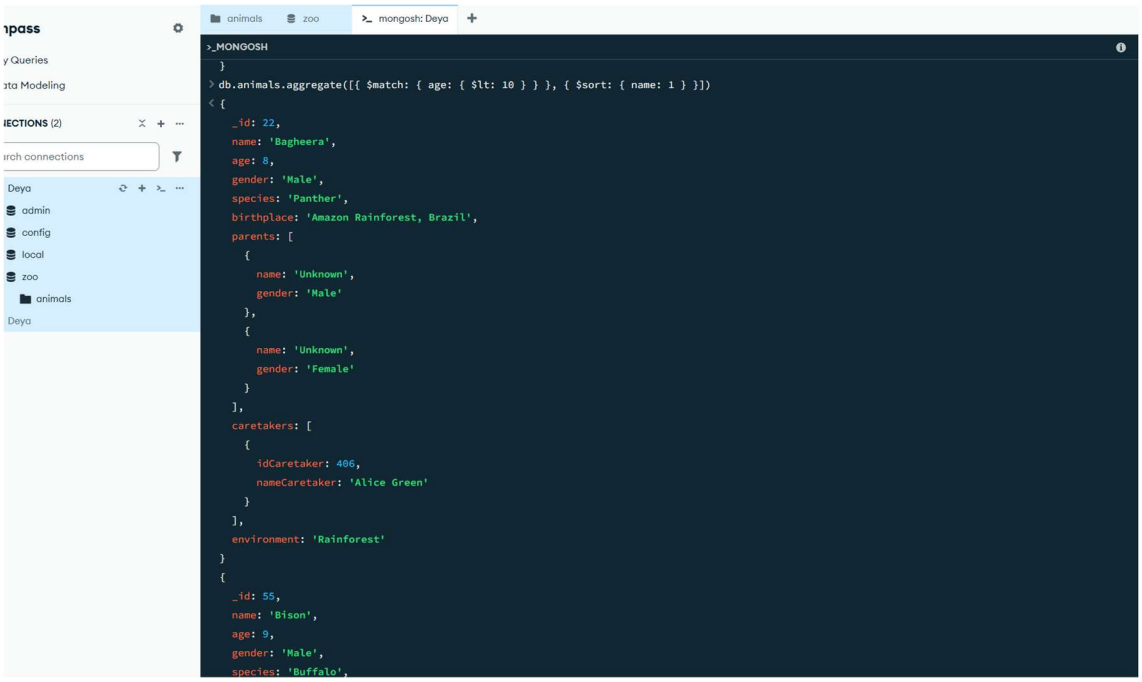
10. Write a MongoDB query to rename the habitat field to environment for all animals in the animals collection.

db.animals.updateMany({}, {animals.updateMany({}, { $rename: {“habitat”: “enviroment”} })



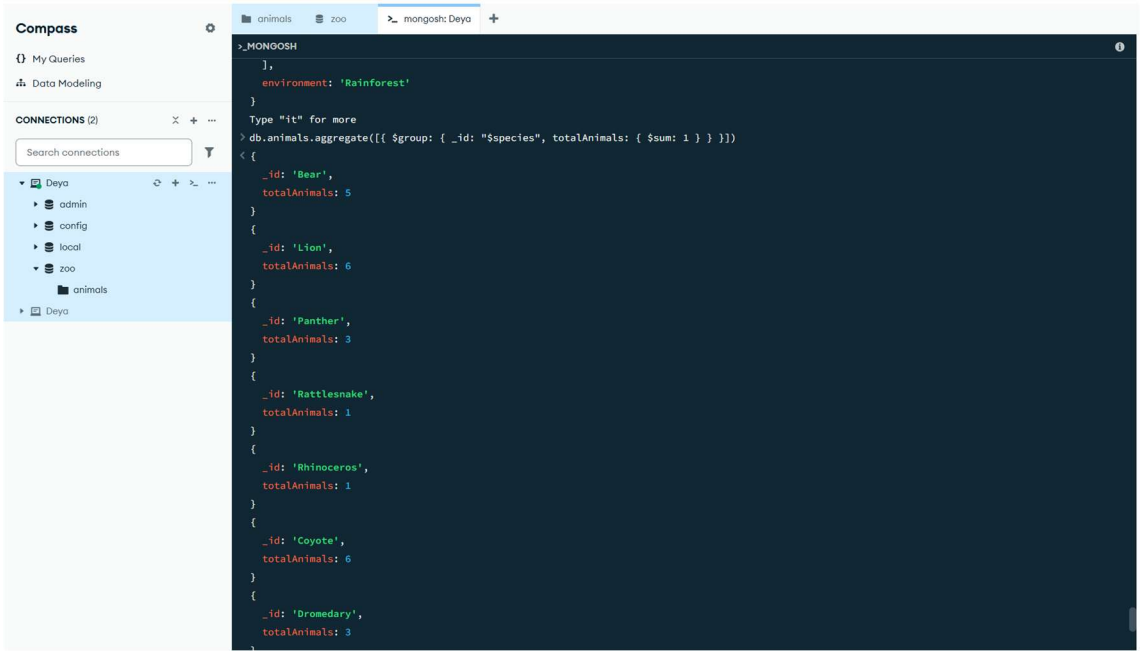
11- Write a MongoDB query using the aggregation framework to find animals aged less than 10 and sort them alphabetically by their name.

db.animals.aggregate([{ $match: { age: { $lt: 10 } } }, { $sort: { name: 1 } }])



12. Write a MongoDB query using the aggregation framework to group animals by their species and count the number of animals in each species.

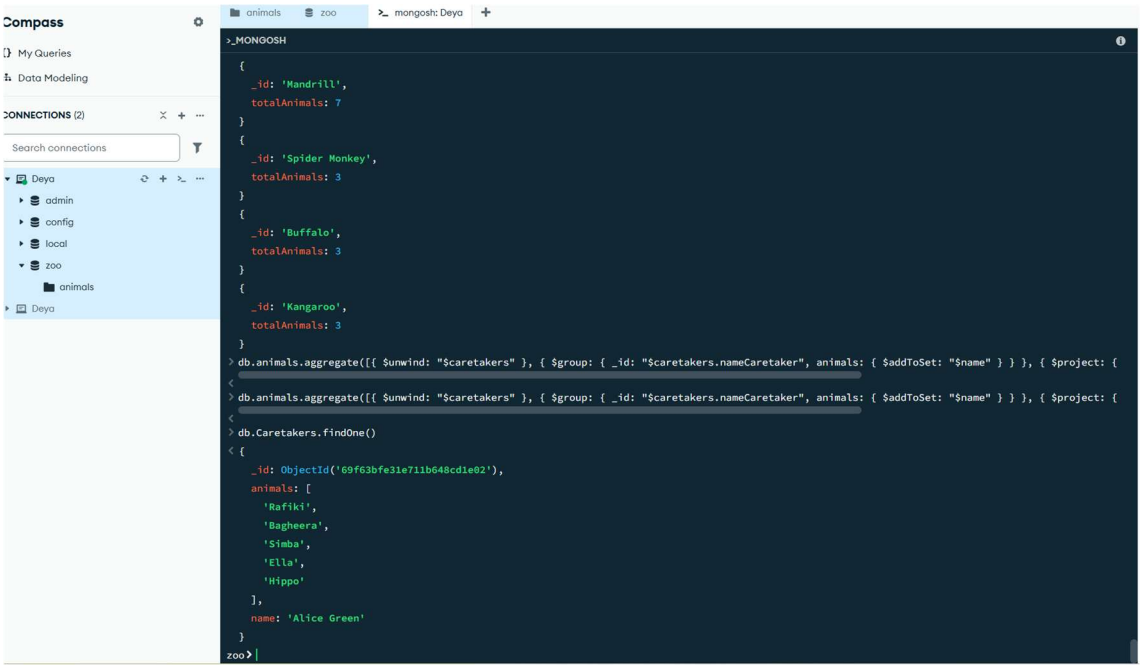
db.animals.aggregate([{ $group: { _id: "$species", totalAnimals: { $sum: 1 } } }])



13. Write a MongoDB query to create a new collection called Caretakers by extracting all caretakers from the animals collection. Ensure that caretakers do not repeat and include an array of the animals they attend to.

db.animals.aggregate([{ $unwind: "$caretakers" }, { $group: { _id: "$caretakers.nameCaretaker", animals: { $addToSet: "$name" } } }, { $project: { _id: 0, name: "$_id", animals: 1 } }, { $out: "Caretakers" }])

db.Caretakers.findOne()

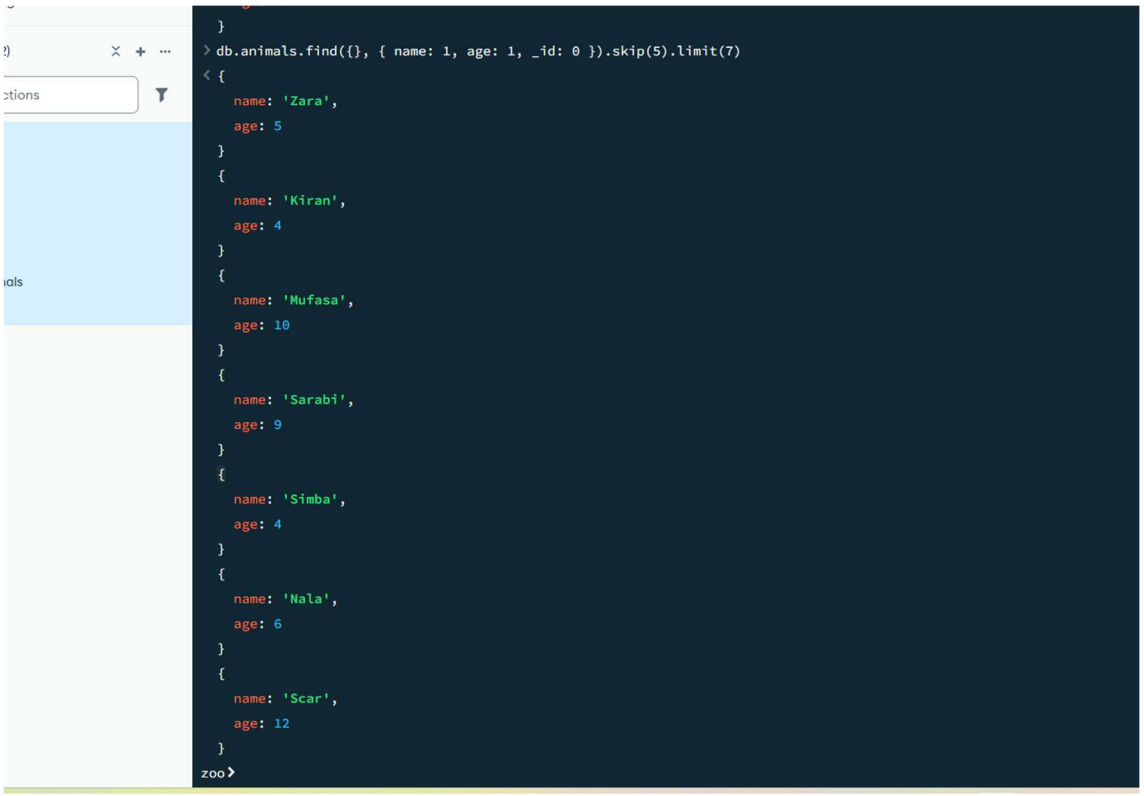


14. Write a MongoDB query to find the top 3 oldest animals.

db.animals.find({}, { name: 1, age: 1, _id: 0 }).sort({ age: -1 }).limit(3)

15.Write a MongoDB query to display animals between the 6th and 12th entries in the collection based on their insertion order.

db.animals.find({}, { name: 1, age: 1, _id: 0 }).skip(5).limit(7)



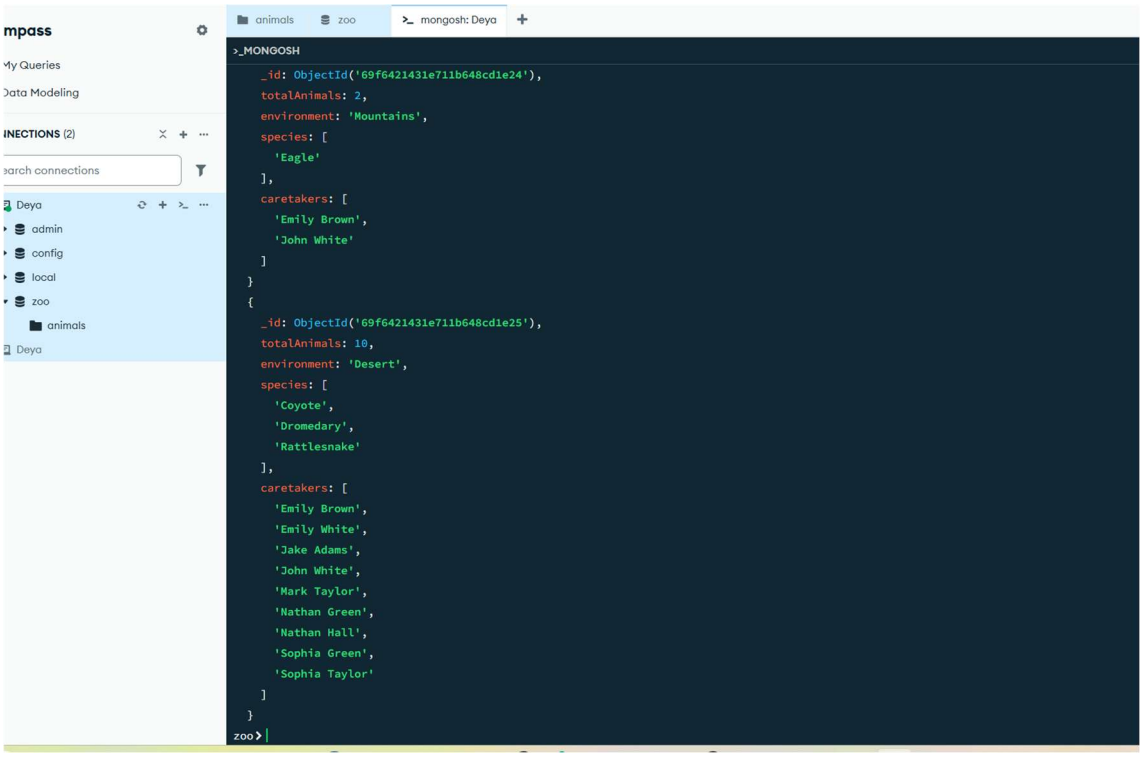
16. Write a MongoDB query using the aggregation framework to create a new collection called Habitats. This collection should group animals by their environment (formerly habitat) and include the following fields.

- The environment name.
- The total number of animals in that environment.
- An array of unique species in that environment, sorted alphabetically.
- An array of unique caretakers who attend animals in that environment.

db.animals.aggregate([{ $unwind: "$caretakers" }, { $group: { _id: "$environment", totalAnimals: { $sum: 1 }, species: { $addToSet: "$species" }, caretakers: { $addToSet: "$caretakers.nameCaretaker" } } }, { $project: { _id: 0, environment: "$_id", totalAnimals: 1, species: { $sortArray: { input: "$species", sortBy: 1 } }, caretakers: { $sortArray: { input: "$caretakers", sortBy: 1 } } } }, { $out: "Habitats" }])

db.Habitats.find()

```
zoo> db.Habitats.find()
[
  {
    _id: ObjectId('679370346cee80a6809e83a7'),
    totalAnimals: 11,
    environment: 'Grasslands',
    species: [ 'Buffalo', 'Elephant', 'Kangaroo', 'Rhinoceros' ],
    caretakers: [
      'Alice Brown',
      'Alice Green',
      'Alice White',
      'Emily Brown',
      'Emily Green',
      'Jake Adams',
      'John Smith',
      'Mark Taylor',
      'Nathan Hall'
    ]
  },
  {
    _id: ObjectId('679370346cee80a6809e83a8'),
    totalAnimals: 10,
    environment: 'Savanna',
    species: [ 'Giraffe', 'Lion' ],
    caretakers: [
      'Alice Green',
      'Emily Brown',
      'Jake Adams',
      'John Smith',
      'Mark Taylor',
      'Michael Davis',
      'Sophia Green'
    ]
  },
  {
    _id: ObjectId('679370346cee80a6809e83a9'),
    totalAnimals: 3,
    environment: 'River',
    species: [ 'Hippopotamus' ],
    caretakers: [ 'Alice Green', 'Mark Brown', 'Sophia Taylor' ]
  },
```


animals
zoo
mongosh: Deya
>_MONGOSH
_id: ObjectId('69f6421431e711b648cd1e24'),
totalAnimals: 2,
environment: 'Mountains',
species: [
'Eagle'
],
caretakers: [
'Emily Brown',
'John White'
]
}
{
_id: ObjectId('69f6421431e711b648cd1e25'),
totalAnimals: 10,
environment: 'Desert',
species: [
'Coyote',
'Dromedary',
'Rattlesnake'
],
caretakers: [
'Emily Brown',
'Emily White',
'Jake Adams',
'John White',
'Mark Taylor',
'Nathan Green',
'Nathan Hall',
'Sophia Green',
'Sophia Taylor'
]
}
zoo
admin
config
local
zoo
animals
Deya
Deya